Evaluating Gossip-Based Consensus in Unreliable IoT Mesh Networks

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1. Problem Statement

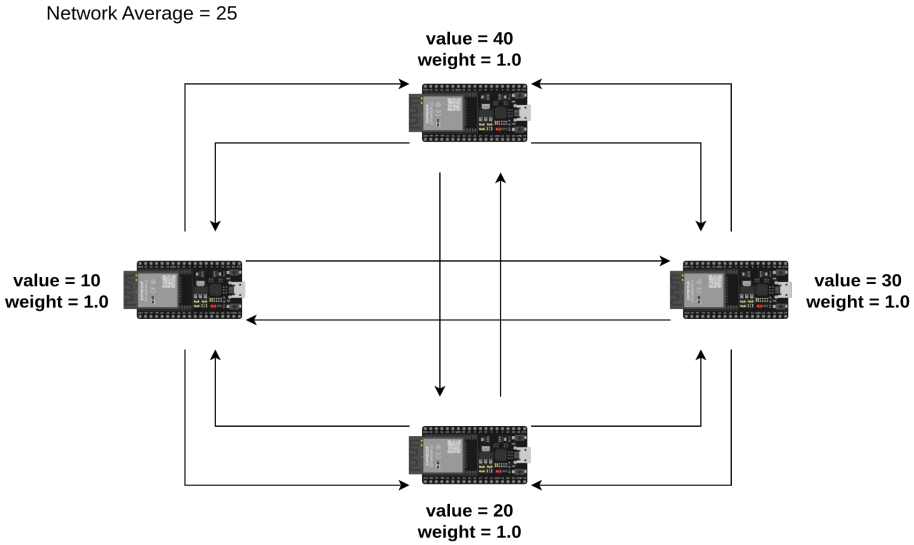
- Decentralized Consensus: IoT nodes must agree on a global state. e.g.
  - Global average temperature to trigger an alarm.
  - Collaborative presence detection.
- Unreliable Communications: Real-world wireless environments are inherently lossy.
- Many high-efficiency consensus protocols (specifically Push-Sum) usually "break" because of this packet loss.
- There is little information on the exact threshold of packet loss where this mathematical drift makes autonomous IoT decisions unreliable.

2. Proposal

- Approach: A comparative empirical study of Push-Sum gossip variants to quantify the trade-off between network efficiency and consensus error.
- Testbed: A physical 4-node mesh using ESP32 communicating via ESP-NOW (WiFi)
- Protocol variants:
  - Naive: Fully asynchronous.
  - Robust: ACK-based handshake.
- Experimental Variable: Software simulated probabilistic packet loss (0, 15, 30, 50 %).
- Outcome: To determine when the overhead of the ACKs becomes necessary for safety.

2. Protocol: Push-Sum

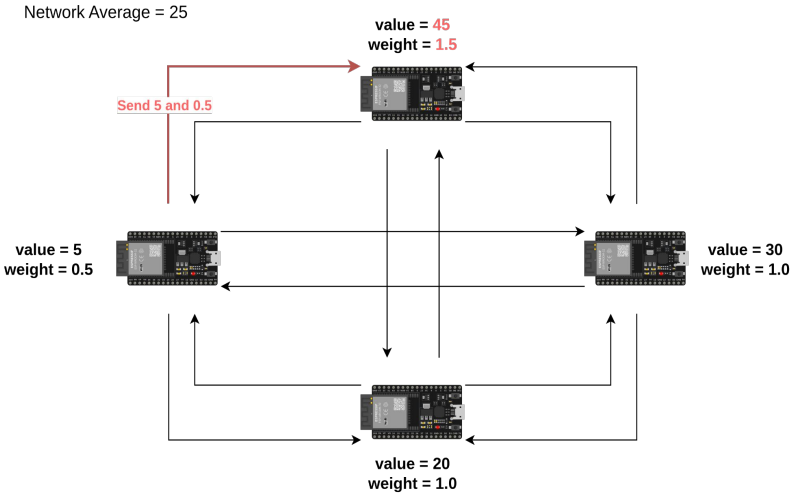
- Each node has a value and a weight.
- The algorithm guarantees that, under optimal conditions, all nodes converge into the same ratio (ratio = value / weight).



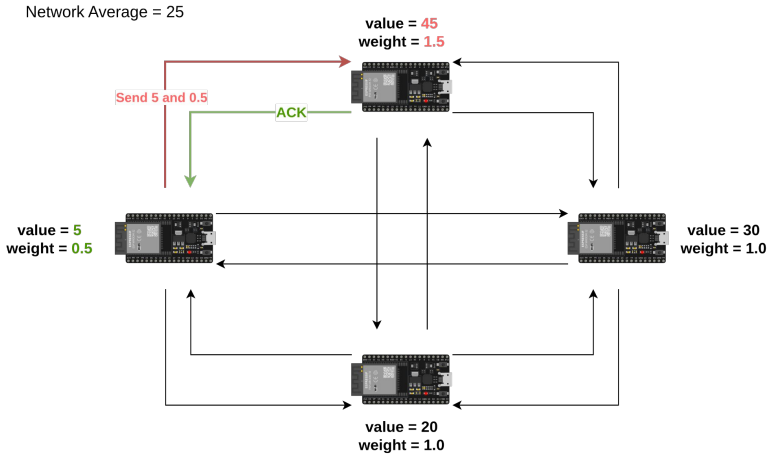
2. Protocol: Push-Sum

- Two variants:

Naive



Robust (ACKs)



3. Hypotheses

Reliability: The Naive protocol will experience significant "mass conservation" errors and value drift as packet loss increases.

Latency: The Robust (ACKs) protocol will maintain mathematical accuracy but will see an exponential increase in time-to-consensus under high loss.

4. Experiments and Results

- **Resources:** 1 developer, 4 ESP32, estimated of 70 hour for project completion.
- **Experimental Design:** Measure the internal ratio (value/weight) at each node and the overall convergence latency under varying simulated packet loss levels.
- **Measure of Success:** Producing a crossover graph that proves exactly when the Robust protocol's latency trade-off becomes a necessary cost to prevent large consensus errors.